

Table 1: Test MSE (mean $\pm$ sample standard deviation over 2 runs). Bracketed values express each method’s mean MSE as a percentage of the formula’s mean MSE; lower is better. Uniform inputs on $[-1, 1]^d$; $N_{\text{train}} = 1000$, $N_{\text{test}} = 2000$. The formula uses $L = 1$, with CV off. MLPs have 3 hidden ReLU layers and use 1001 full-batch Adam updates at learning rate 0.001. SN and orthogonal MLP bounds are 1.

(a) Hard

Target	d	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
64 signed bumps	2	$(2.235 \pm 0.177) \times 10^{-4}$	$(1.048 \pm 0.323) \times 10^{-3}$	$(1.718 \pm 0.070) \times 10^{-3}$	$(1.643 \pm 0.017) \times 10^{-3}$	$(2.062 \pm 0.015) \times 10^{-3}$
		[100.0%]	[468.7%]	[768.4%]	[735.1%]	[922.6%]
Folding: 4 layers	2	$(9.119 \pm 3.013) \times 10^{-5}$	$(3.415 \pm 0.499) \times 10^{-4}$	$(9.302 \pm 0.791) \times 10^{-3}$	$(6.562 \pm 1.174) \times 10^{-3}$	$(6.231 \pm 1.164) \times 10^{-2}$
		[100.0%]	[374.5%]	[10200.5%]	[7195.6%]	[68335.9%]

(b) Tied

Target	d	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
Folding: 1 layer	100	$(1.315 \pm 0.055) \times 10^{-1}$	$(1.404 \pm 0.107) \times 10^{-1}$	$(1.234 \pm 0.055) \times 10^{-1}$	$(1.261 \pm 0.046) \times 10^{-1}$	$(1.200 \pm 0.037) \times 10^{-1}$
		[100.0%]	[106.7%]	[93.8%]	[95.8%]	[91.3%]
Folding: 4 layers	100	$(5.325 \pm 0.311) \times 10^{-2}$	$(5.021 \pm 0.262) \times 10^{-2}$	$(4.918 \pm 0.256) \times 10^{-2}$	$(4.781 \pm 0.281) \times 10^{-2}$	$(4.221 \pm 0.268) \times 10^{-2}$
		[100.0%]	[94.3%]	[92.4%]	[89.8%]	[79.3%]

(c) MLP exploits different inductive bias

Target	d	Formula	ReLU MLP	SN MLP	Orthogonal MLP	Constant
Affine, $\ w\ _1 = 1$	100	$(5.672 \pm 0.137) \times 10^{-3}$	$(2.863 \pm 0.155) \times 10^{-4}$	$(1.651 \pm 0.215) \times 10^{-4}$	$(2.817 \pm 0.073) \times 10^{-3}$	$(5.260 \pm 0.018) \times 10^{-3}$
		[100.0%]	[5.0%]	[2.9%]	[49.7%]	[92.7%]
Tanh ridge	100	$(1.475 \pm 0.228) \times 10^{-1}$	$(1.148 \pm 0.242) \times 10^{-2}$	$(1.190 \pm 0.061) \times 10^{-2}$	$(6.165 \pm 1.605) \times 10^{-2}$	$(1.725 \pm 0.421) \times 10^{-1}$
		[100.0%]	[7.8%]	[8.1%]	[41.8%]	[117.0%]

Notes. $d = 2$: $P_{\text{formula}} = 3001$, $P_{\text{MLP}} = 2961$ (ratio 0.9867); $d = 100$: $P_{\text{formula}} = 101001$, $P_{\text{MLP}} = 100801$ (ratio 0.9980). All three MLP families match these stored-scalar counts. The constant predictor stores the training-label mean. The panel labels denote the expected behavior; a measured tie means all three MLP percentages lie in [85%, 115%]. The affine target has Euclidean Lipschitz constant $\|w\|_2 < 1$, whereas the tanh target uses $\|w\|_2 = 1$ and an interior zero crossing.